

Error Mitigation in Deep Parameterized Quantum Circuits

Panos Constantinides¹, Spyros Tserkis^{1,2} and Dimitris G. Angelakis^{3,2,4}

¹School of Electrical and Computer Engineering, Technical University of Crete, Chania 73100, Greece

²Institute of Quantum Computing and Quantum Technologies, NCSR Demokritos, Greece

³School of Electronics and Computer Science, University of Southampton, Southampton SO17 1BJ, UK

⁴Center for Quantum Technologies, National University of Singapore, 3 Science Drive 2, 117543, Singapore



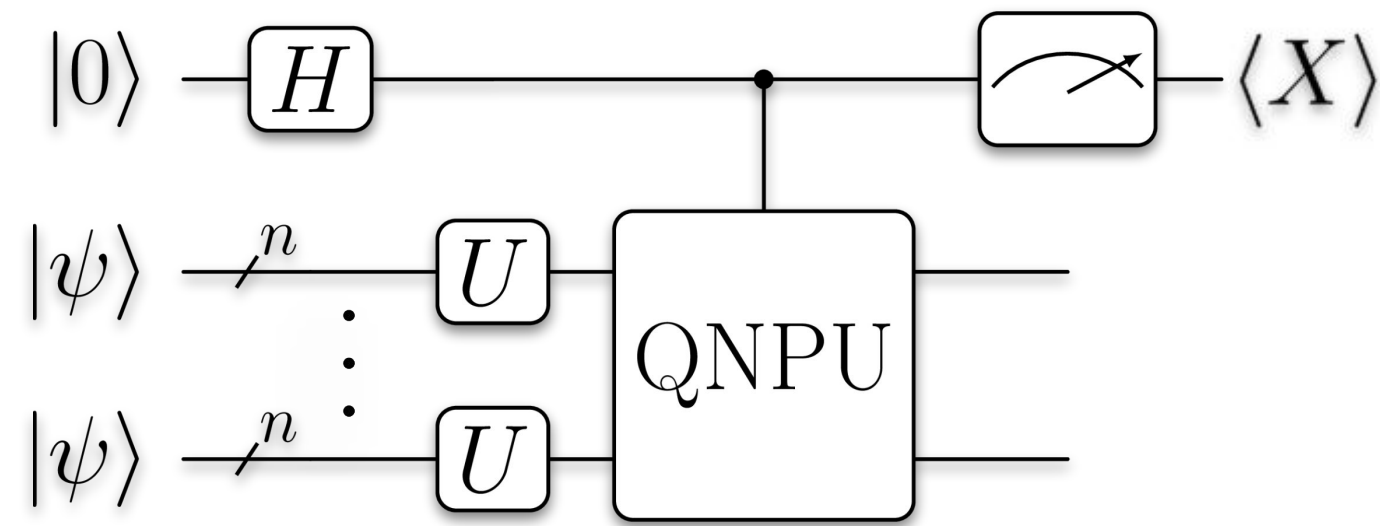
Abstract

This poster presents results of an ongoing undergraduate research project. The main focus of this work is error mitigation in parameterized Hadamard test circuits using zero-noise extrapolation (ZNE) and echo verification (EV). These include ansatz circuits relevant to variational quantum algorithms in quantum computational fluid dynamics (QCFD), such as the solution of the Burgers' equation. We analyze both unitary and dynamical (non-unitary) ansatz circuits and conclude that ZNE consistently outperforms EV in the studied regime in terms of absolute error, although it suffers from larger standard deviations. We also observe that dynamical circuits increase the standard deviation and perform slightly worse in terms of removing errors compared to their unitary counterparts. Finally, we find that the combination of the two error mitigation methods is more favorable for the dynamical circuits, at least in the studied case.

Error Mitigation & Variational Techniques

Variational Quantum Algorithms

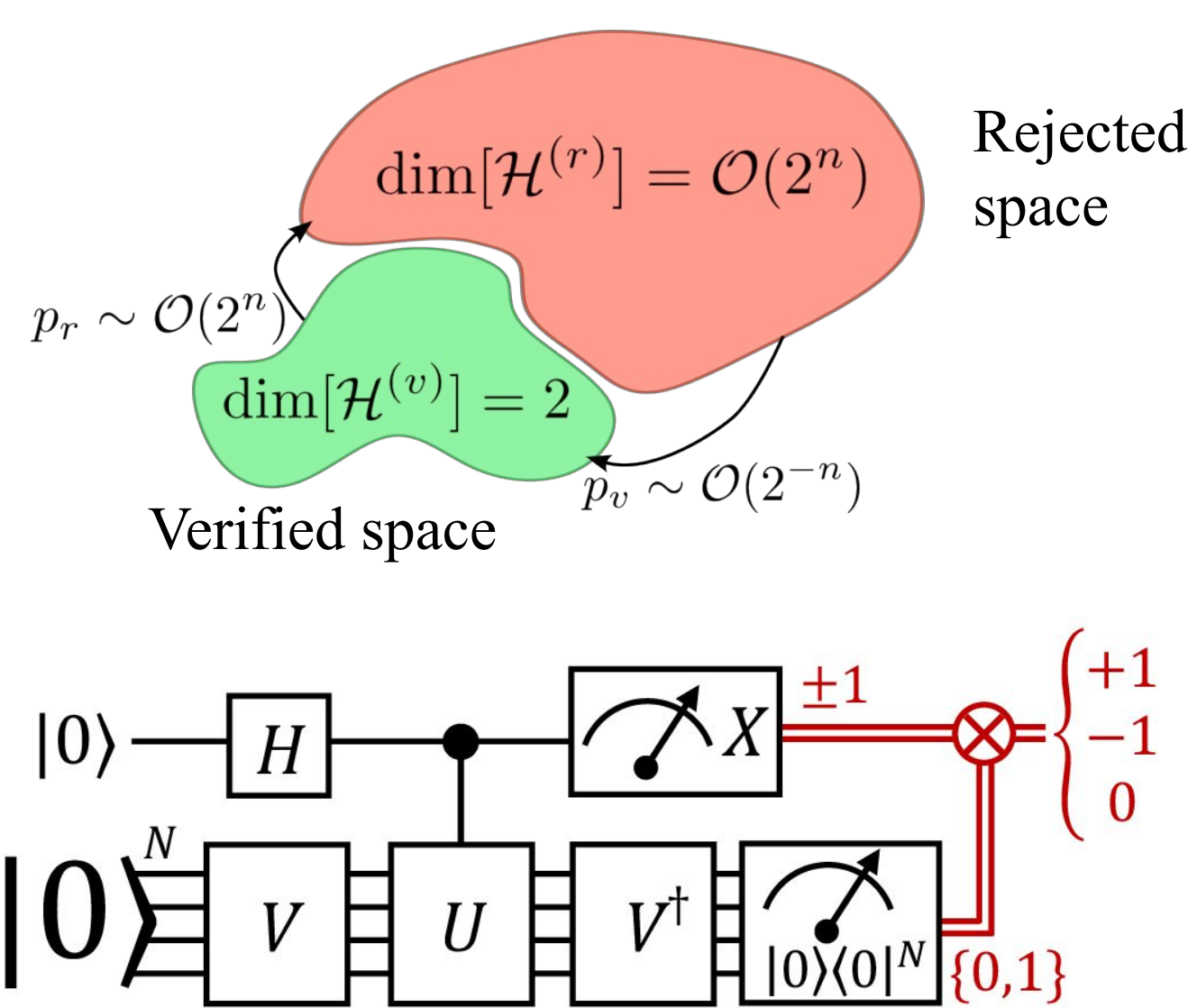
Variational quantum algorithms [1] rely on parameterized quantum circuits, known as ansatz circuits, designed to prepare quantum states that approximate the solution to a given problem. These circuits typically consist of alternating layers of single qubit rotations and entangling two-qubit gates, with the parameters updated iteratively by a classical optimizer to minimize a cost function dependent on the problem.



Goal: In practice quantum computation suffers from device noise, thus we employ error mitigation techniques to reduce noise.

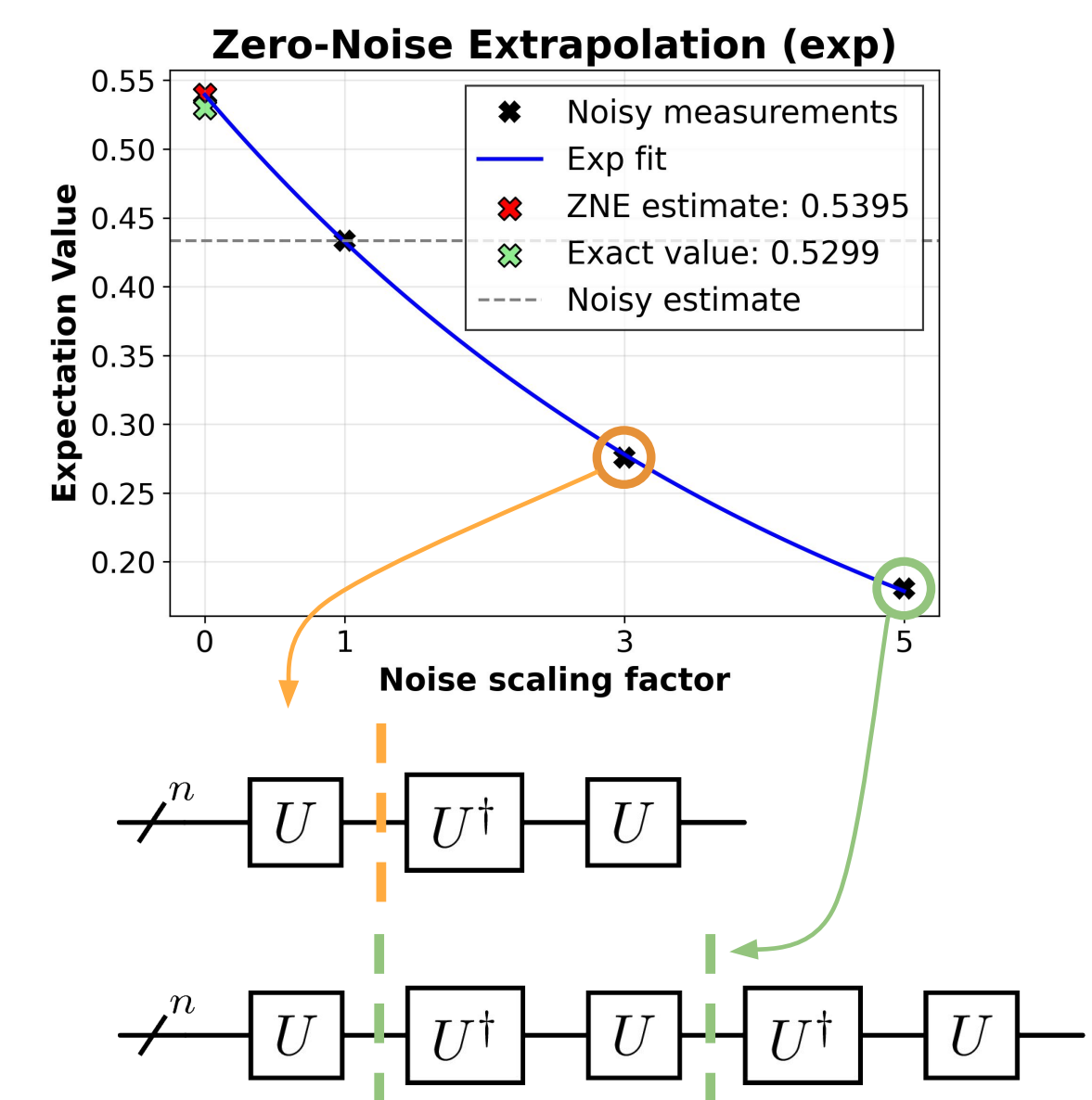
Echo Verification

1. Prepare the register with V and apply C-U
2. Measure register qubits
3. Post-select based on register measurements



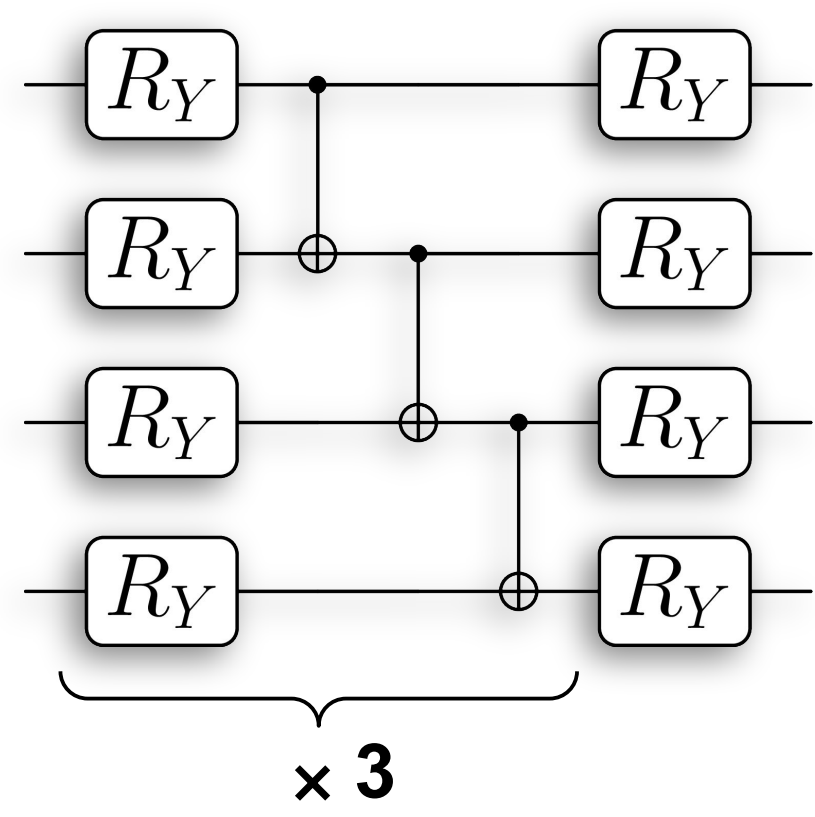
Zero-Noisy Extrapolation

1. Amplify noise in a well-defined way
2. Extrapolate in post-processing
3. Zero-noisy estimate via mean-squared error fitting

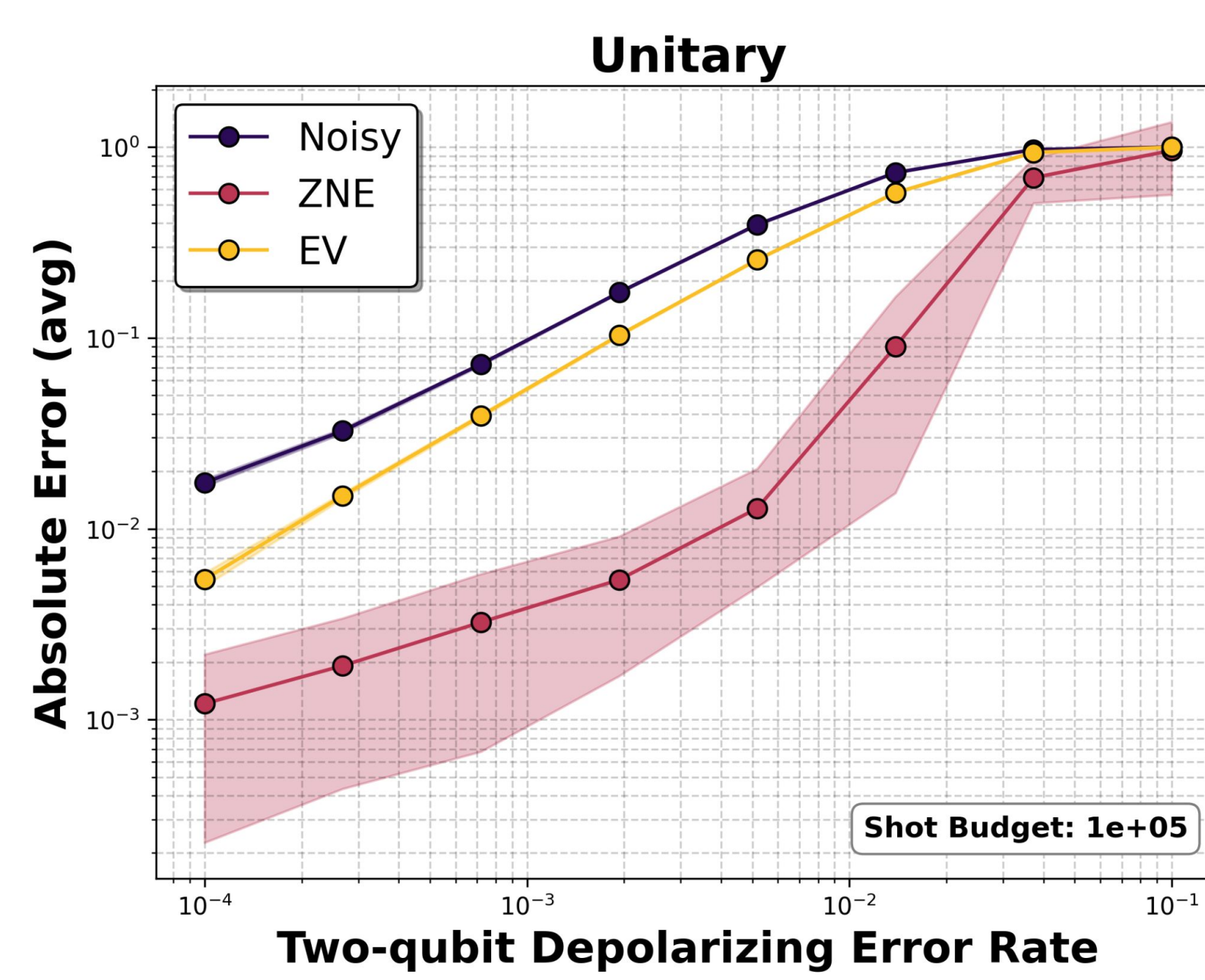


Results & Conclusions

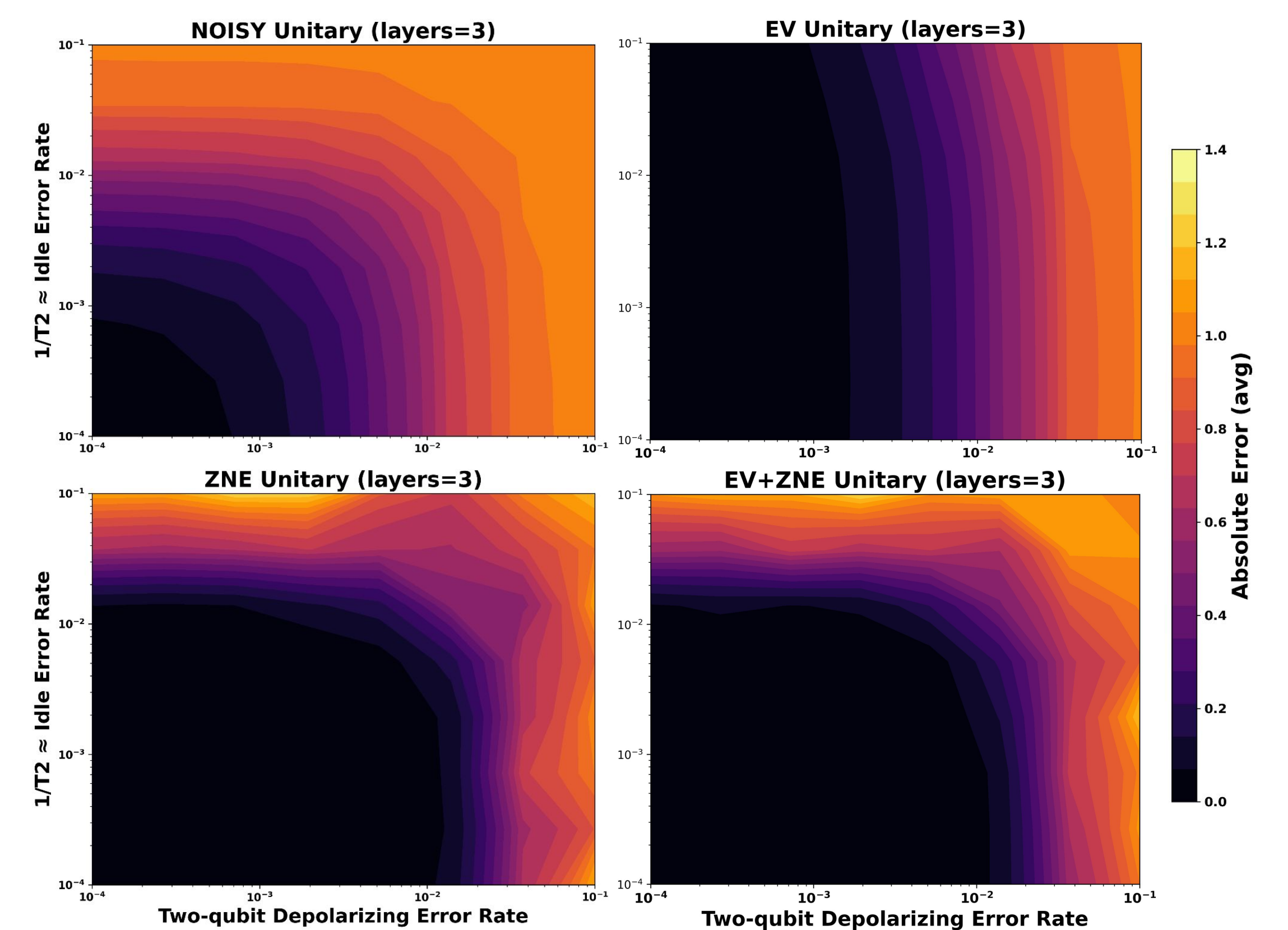
Unitary Ansatz



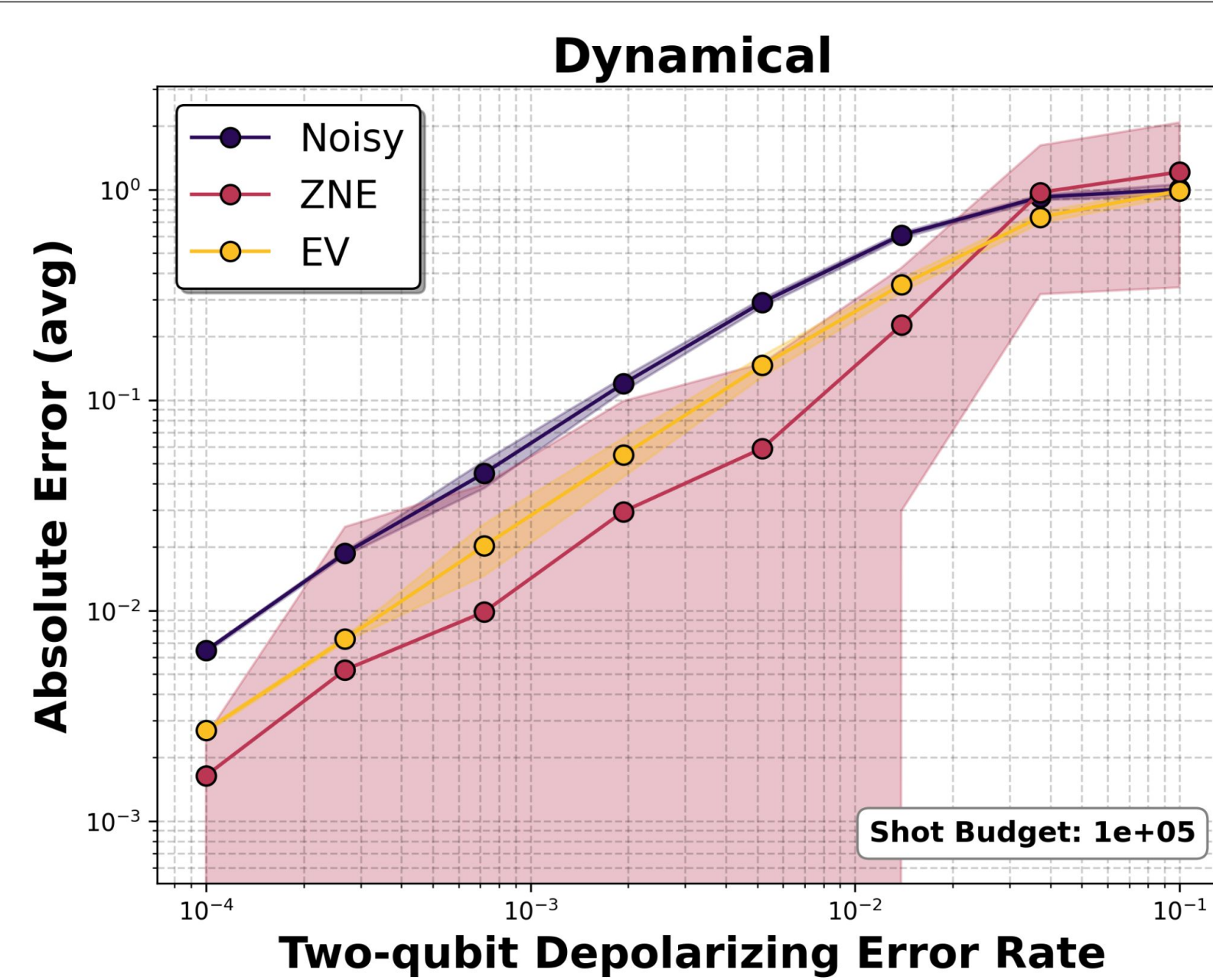
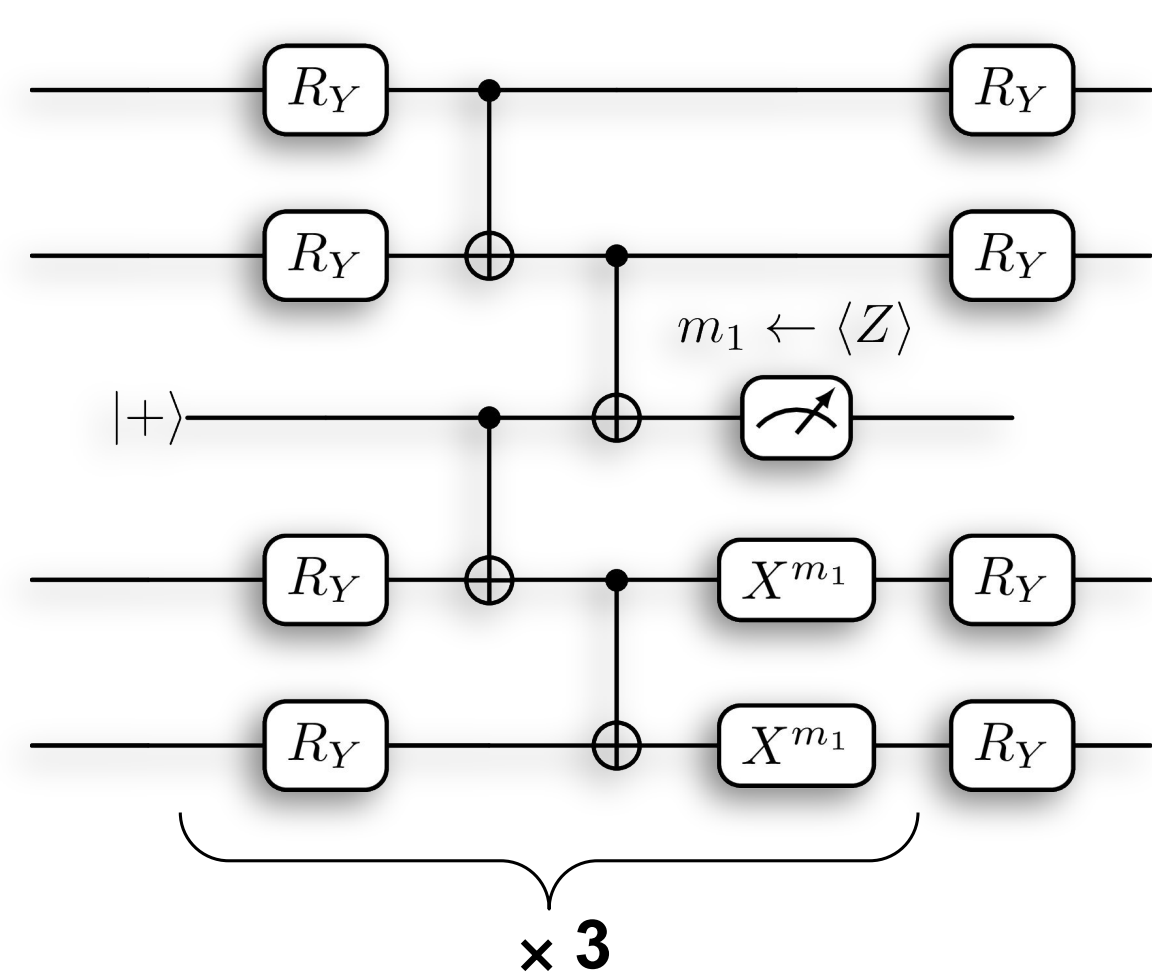
Target expectation value = 1.0



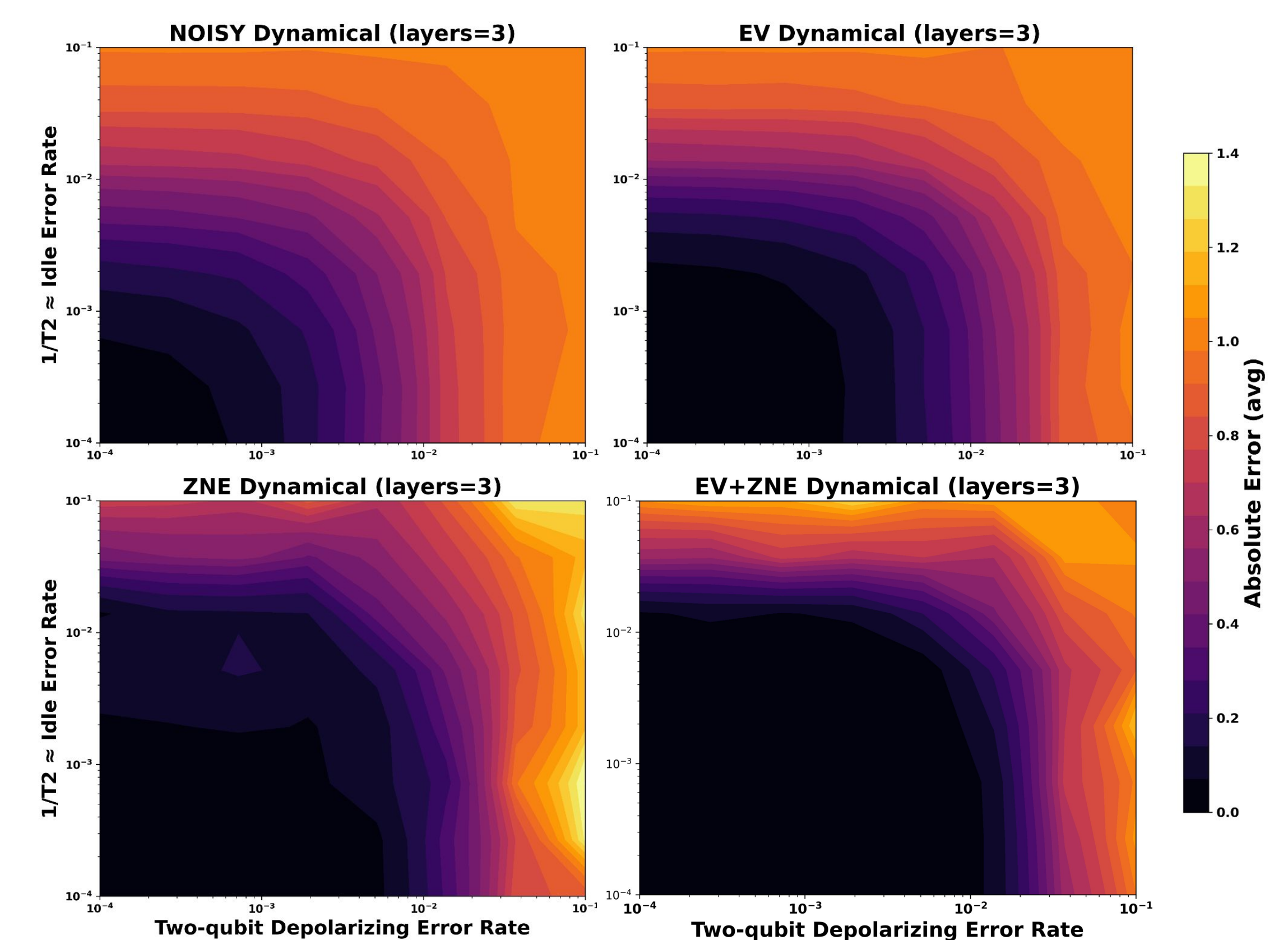
- ZNE is consistently better than EV in terms of absolute error, but it suffers from higher standard deviations due to the instability of extrapolation.
- The combination EV and ZNE does not provide any significant advantage in the case of 3 layers and 4 qubits, although it improves compared to ZNE in terms of stability.



Dynamical Ansatz

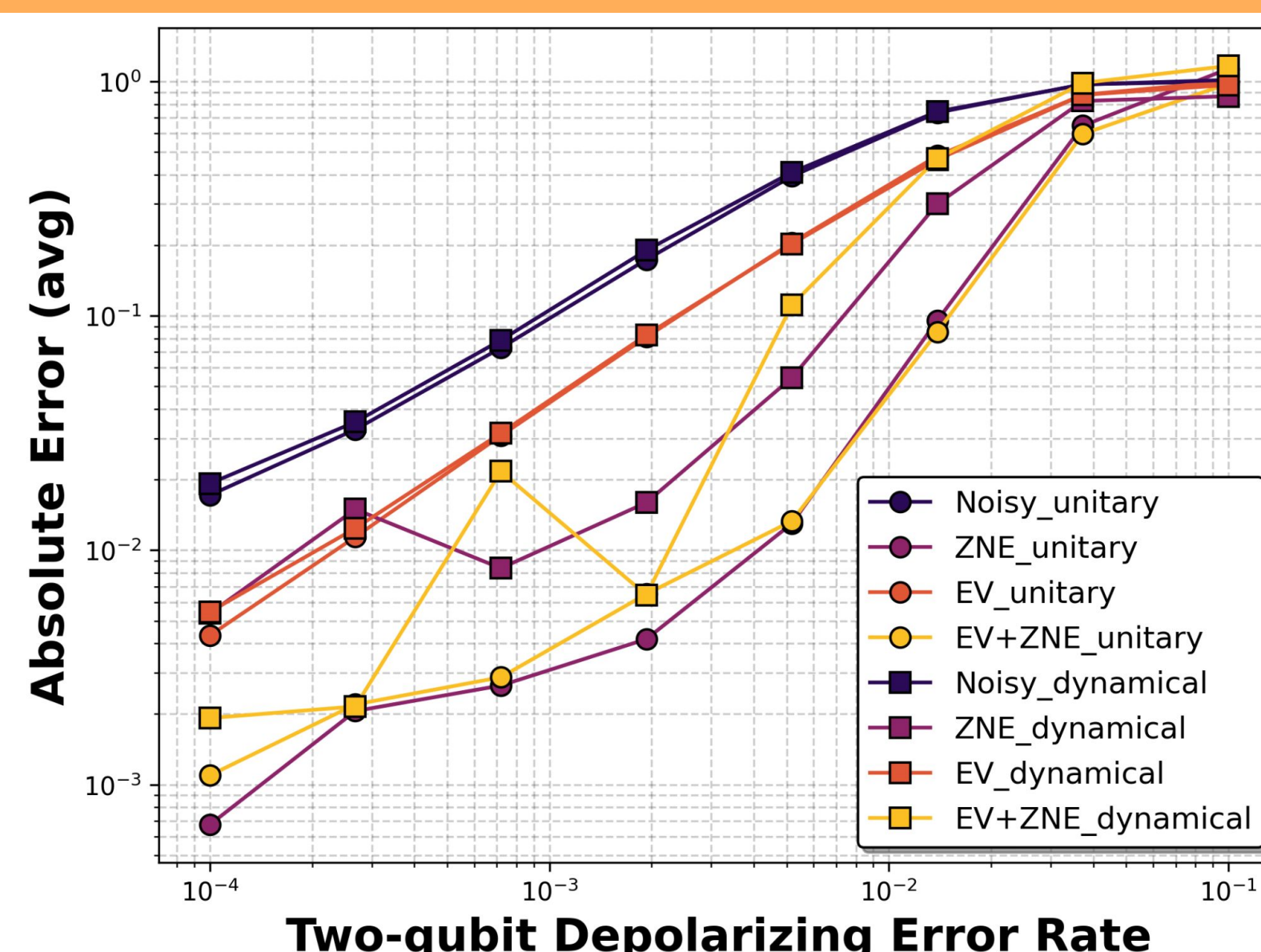


- Dynamical circuits [4] achieve constant depth at the expense of ancilla qubits, mid-circuit measurements and classically controlled gates.
- Within the studied regime, dynamical circuits demonstrate higher variance due to the larger number of gates, we expect this to change for larger number of qubits.



Future work

1. Include more sophisticated error mitigation techniques [5].
2. Reduce the dependency of QEM methods on the noiseless (ideal) value.
3. Apply error mitigation techniques in more complex QNPU circuits relevant to variational quantum algorithms.
4. Optimize Burgers' equation through error mitigation by achieving target accuracy with fewer layers.



References

1. M. Cerezo et al., Variational quantum algorithms, Nature Reviews Physics 3, 625–644 (2021)
2. T. O'Brien et al., Error Mitigation via Verified Phase Estimation, PRX Quantum 2, 020317, (2021)
3. S. Polla et al., Optimizing the information extracted by a single qubit measurement, Phys. Rev. A 108, 012403, (2023)
4. S. Tserkis et al., Depth optimization of CNOT ladder circuits, arXiv:2511.13256, (2026)
5. J. Koh et al., Readout Error Mitigation for Mid-Circuit Measurements and Feedforward, PRX Quantum 7, 010317, (2026)